# Data Mining & Business Intelligence (IT647)
# Selected Topics In Data Science (IT688)

Information Technology Department
College Of Computer
Qassim University

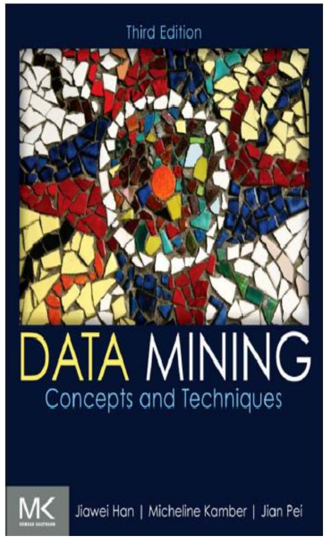
DR.MONA ALSALEEM

# Data Mining & Business Intelligence Course_IT674

# Data Mining

## Business Intelligence Introduction

Text Book: “Data Mining: Concepts and Techniques”, Third Edition



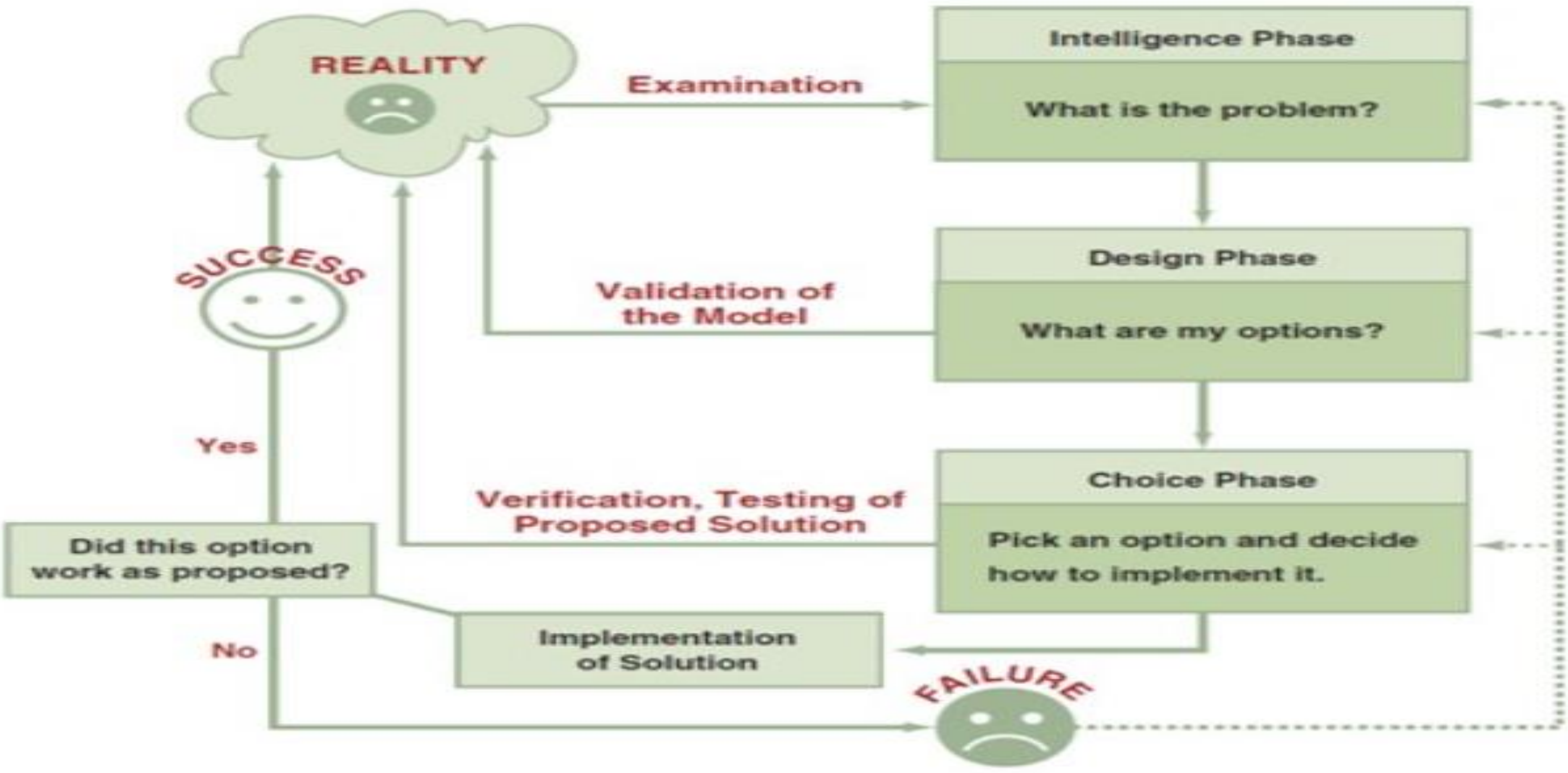
# Outlines

- Managers and Decision Making
- What Is Business Intelligence .
- Why Business Intelligence? Benefits.
- BI Application Areas
- Business Intelligence Life Cycle
- A Framework for Business Intelligence
- Data extraction and transformation in Business Intelligence (ELT)

# The Manager's Job and Decision Making

- Management
  - a process by which an organization achieves its goals through the use of resources (people, money, materials and information)
- Three Basic Roles of Managers (Mintzberg , 1973)
  1. Interpersonal roles (rel-based): leader
  2. Informational roles (comm-based): monitor, spokesperson, analyzer
  3. Decisional roles (action-based): entrepreneur, resource allocator, negotiator

# Process and Phases in Decision Making



# Why Managers Need IT Support ?

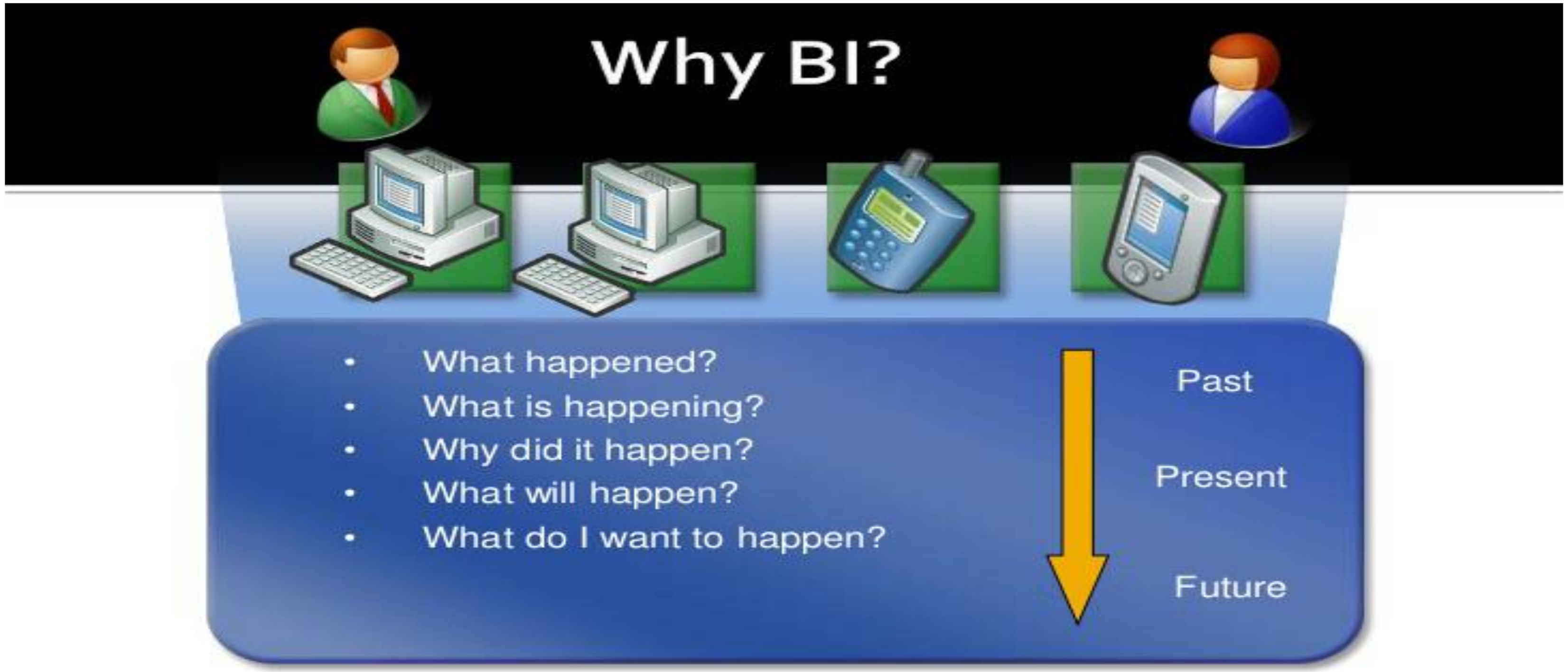
- The number of alternatives is constantly increasing.
- Most decisions are made under time constraints;
- Uncertainty in the decision environment
- Group decision making required

# What Is Business Intelligence

- **Business Intelligence**
  - **Is a broad category of <u>applications</u>, <u>technologies</u> and <u>processes</u> for gathering, storing, accessing and analyzing data to help business users make better decisions.**
- **BI applications enable decision makers to quickly ascertain the status of a business enterprise by examining key information.**

# Why Business Intelligence?

- **BI is for answering following business related questions technically**



# Benefits of Business Intelligence

- **Improve Management Processes – planning, controlling, measuring and/or changing results in increased revenues and reduced costs;**
- **Improve Operational Processes – fraud detection, order processing;**
- **Better Adjustment settings – Competitor analysis, adjustments settings to changing trends;**
- **Predict the Future – Predictive analysis, Forecasting.**

# BI Application Areas

- BI can be applied in all “businesses” both private and public sectors
- Private
  - Retail, manufacture, real-estate, sports, media, publication, etc.
- Public (non-profit)
  - Education, government, healthcare, association, etc .

# BI Applications for Data Analysis

- **Multidimensional Analysis or Online Analytical Processing**
- **Data Mining**
- **Decision Support Systems**

# Multidimensional Analysis or Online Analytical Processing

- **Multi-dimensional Analysis**
  - Multi-dimensional analysis is a data analysis approach that examines information across multiple dimensions at the same time, such as time, location, product, and customer
- **Online Analytical Processing**
  - OLAP involves "slicing" and "dicing" data stored in a dimensional format, drilling down in the data to greater detail and aggregating the data

# Data Mining

- **Data Mining: the process of searching for valuable business information in a large database, data warehouse, or data mart**
- **Two Basic Data Mining Operations**
  - **Predicting trends and behaviors**
  - **Identifying previously unknown patterns**

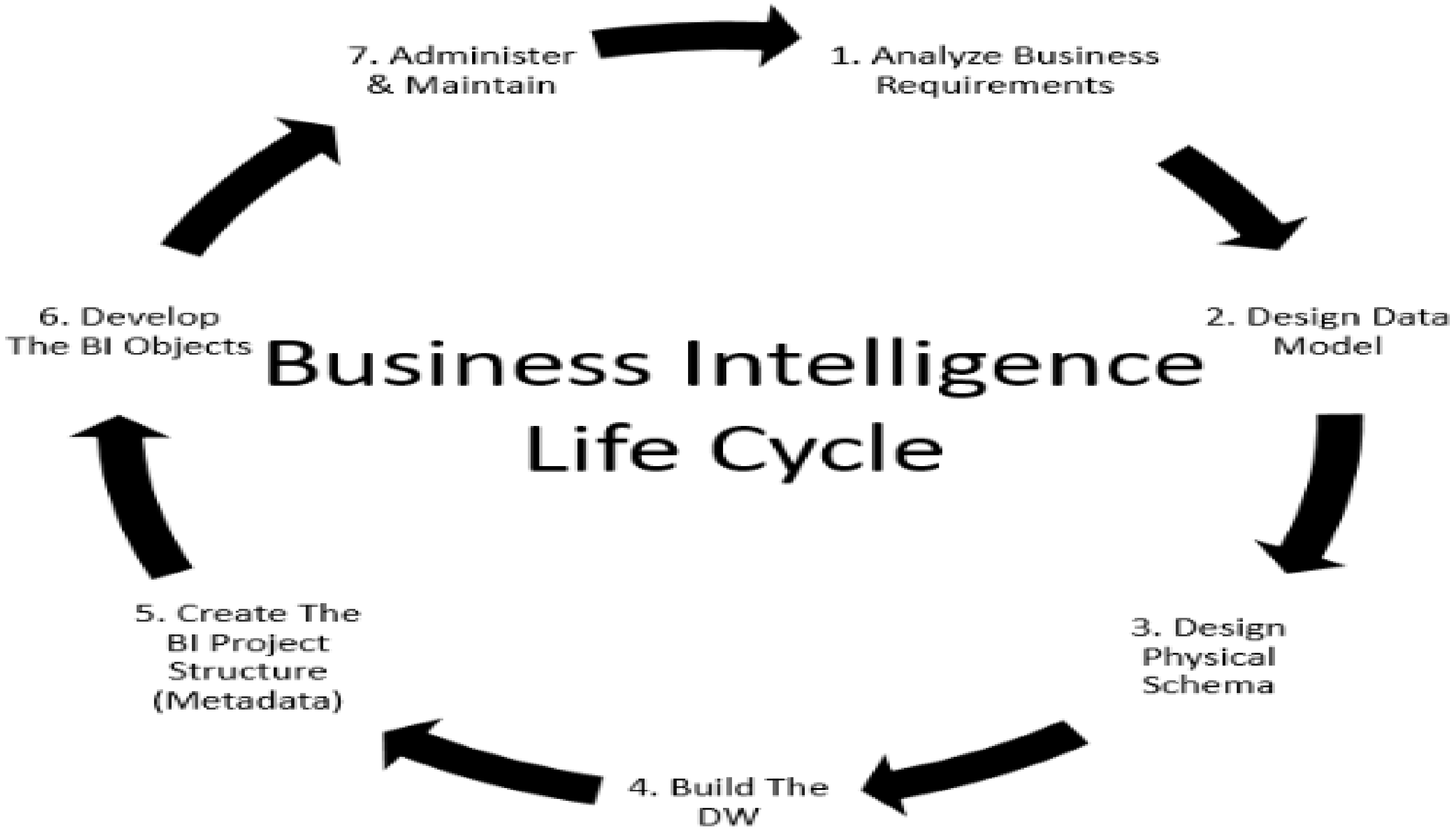
# BI Applications for Presenting Results

- **Dashboards**
  - Provides easy access to timely information and direct access to management reports.
  - They evolved from executive information systems, which were designed specifically for the information needs of top executives.
- **Data Visualization Technologies**
  - Data presented to users in visual formats such as text, graphics and tables following data processing.
  - Data Visualization makes IT applications more attractive and understandable to users
- **Real-Time BI**
  - Includes the use of real time data for analysis as it is created rather than using historical data for analysis.

# The Capabilities of Dashboards

| Capability | Description |
| --- | --- |
| Drill down | The ability to go to details, at several levels, it can be done by a series of menus or by clicking on a drillable portion of the screen |
| Critical Success Factors (CSFs) | The factors most critical for the success of business. These can be organizational, industry, departmental, for individual workers. |
| Key Performance Indicators (KPI) | The specific measures of CSFs |
| Status access | The latest data available on KPI or some other metrics, often in real time. |
| Trend analysis | Short, medium and long term trend of KPIs or metrics, which are projected using forecasting methods. |
| Exception reporting | Reports that highlight deviations larger than certain thresholds, Reports may include only deviations. |

# Business Intelligence Life Cycle



# Business Intelligence Life Cycle

1. **Analyze Business Requirements** - reviewing business requirements to determine the types of analysis user need to perform.

2. **Design Data Model** - Based on the business requirements, design the logical data model, which shows the information that users want to analyze and the relationships that exists within the data.

3. **Design the Physical Schema** - Using the data model design physical schema (creating dimension and fact table hence star schema) which defines the content and structure of the data warehouse.

4. **Build the Data Warehouse** - Build the data warehouse according to the schema design and load data into the warehouse.

# Business Intelligence Life Cycle

5. **Create the Project Structure (Metadata)** - **Create the metadata and begin to connect and map the metadata to table in the data warehouse e.g.**

6. **Develop The BI Objects** - **Develop object, like reports, scorecards and dashboard.**

7. **Administer and Maintain the Project** - **Administer and maintain the project as it undergoes continued development and changes, monitor performance and make adjustments to improve it, manage security, and perform other ongoing administrative tasks.**

# Data Extraction and Transformation in BI

- ETL is a process used in Business Intelligence (BI) to extract, transform, and load data into a data warehouse or other data repository.
- This process is essential for making data available for analysis and decision-making.

# Data Extraction and Transformation in BI

## Data Extraction

- Data extraction involves retrieving data from various sources, such as databases, APIs, or flat files.
- This data may be structured or unstructured, and may come from internal or external sources.

## Data Transformation

- Data transformation involves converting the data into a format that is suitable for analysis.
- This may involve cleaning and filtering the data, as well as performing calculations or aggregations.

## Data Quality Assurance

- Data quality assurance involves ensuring that the data is accurate, complete, and consistent.
- This may involve validating the data against business rules or performing data profiling to identify anomalies or outliers.

# Data Extraction and Transformation in BI

## Data Integration

- Data integration involves combining the data from different sources into a single data repository.
- This may involve mapping data elements between different systems or resolving conflicts between data values.

## Data Visualization

- Data visualization involves presenting the data in a graphical format that is easy to understand and interpret.
- This may involve creating charts, graphs, or dashboards that highlight key insights or trends.

# Data Extraction

- Sources of Data
  - Data can be extracted from various sources, including databases, APIs, spreadsheets, and flat files.
- Extracting Data from Databases
  - Data can be extracted from databases using SQL queries or database connectors.
- Extracting Data from APIs
  - Data can be extracted from APIs using API connectors or custom scripts.

# Data Transformation

## Data Cleaning

- Data cleaning is the process of identifying and correcting or removing errors, inconsistencies, and inaccuracies from data.

## Data Transformation Techniques

- Data transformation involves converting data from its original format into a format that is more suitable for analysis.

## Data Enrichment

- Data enrichment is the process of enhancing data with additional information from external sources.
- This step can include adding demographic or geographic data to customer records, or appending social media data to product sales data.

# Data Quality Assurance

## Data Profiling

- Data profiling is the process of analyzing data to identify and understand its structure, content, and relationships. This helps to ensure that data is accurate, complete, and consistent.

## Data Cleansing

- Data cleansing involves identifying and correcting or removing errors, inconsistencies, and redundancies in data. This is important for ensuring that data is accurate and reliable.

## Data Validation

- Data validation is the process of ensuring that data meets certain criteria or rules. This helps to ensure that data is accurate and consistent, and can be relied upon for decision-making.

# Data Integration

## ETL vs ELT

- In traditional ETL processes, data is extracted from source systems, transformed to fit the target system, and then loaded into the target system.
- In ELT, the data is first extracted and loaded into the target system, and then the transformation occurs within the target system.

## Data Warehouse Integration

- Data from disparate sources can be integrated into a data warehouse using ELT processes. This allows for a single source of truth for reporting and analysis.

## API Integration

- Application Programming Interfaces (APIs) can be used to integrate data from external sources into a target system using ELT. This allows for real-time data integration and reduces the need for manual data entry.

# ETL Tool Overview: Talend

## ETL Process

- The ETL (Extract, Transform, Load) process is a critical component of data management. It involves extracting data from various sources, transforming it into a format suitable for analysis, and loading it into a data warehouse or other target system.

## Talend in the ETL Process

- Talend is an open-source ETL tool that provides a comprehensive solution for data integration.
- It supports a wide range of data sources, including databases, cloud services, and file systems, and offers a variety of transformation options to customize the data flow.

# ETL Tool Overview: Talend

## Data Extraction

- **Talend provides a variety of connectors for extracting data from various sources, including JDBC, ODBC, and HTTP. It also supports real-time data extraction from streaming sources, such as Kafka and Apache Kafka.**

## Data Transformation

- **Talend offers a wide range of transformation options, including SQL, Java, and Python. It also supports data profiling and data quality checks to ensure the accuracy and consistency of the data.**

## Data Loading

- **Talend supports a variety of data loading options, including batch loading, incremental loading, and real-time loading. I**

# Summary

- ***Business Intelligence* - the processes, technologies, and tools that help the organization change data into information, information into knowledge and knowledge into plans that guide the organization.**

- ***Technologies for gathering, storing, analyzing and providing access to data to help enterprise make better business decision.***

- ***Business Intelligence Applications for Data Analysis (Online Analytical Processing).***

- ***Business Intelligence Life Cycle - A life cycle oriented BI success model is presented including an outline of the constructs, their relationships, and associated measurement items.***

- ***ETL is a process used in Business Intelligence (BI) which essential for making data available for analysis and decision-making.***

# Thank you